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Associations between depression and parenting of fathers in Head Start and parental self-efficacy as a protective factor

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ABSTRACT

Head Start/Early Head Start (HS/EHS) programs deliver comprehensive early childhood services in the U.S. to support low-income families to improve a range of early childhood outcomes and to develop positive parenting and parent-child relationships. From the beginning of the program, increasing fathers' involvement with their children in HS/EHS has been an important emphasis. However, depressive symptoms among low-income fathers can negatively impact their relationships with their children. Protective factors such as parental self-efficacy (PSE) can potentially buffer the effects of depression. In this study, we aimed to understand the associations between low-income HS/EHS fathers' depressive symptoms and relationship closeness and conflict with their young children aged 5 and below. We also examined if the PSE of fathers can buffer the negative associations between depression and father-child relationship. We analyzed cross-sectional data collected from fathers ($n = 102$) who participated in a parenting support program while receiving HS/EHS services in the United States. Participants were primarily African American (83%), of mean age 32 years, and lived with their children (74%). Using multiple linear regression, we found that fathers' depression was significantly associated with father-child closeness ($B = -0.37$, $p < .001$) and conflict ($B = 0.28$, $p < .05$). More importantly, PSE functioned as a protective factor by reducing the negative associations between depression and fathers' relationship closeness and conflict with their children. Implications include providing mental health assessments and support to low-income fathers. HS/EHS services or early childhood services that aim to promote positive parenting practices and develop healthy parent-child relationships among low-income fathers should give attention to strengthening PSE in fathers.

1. Introduction

Since 1995, the Head Start/Early Head Start (HS/EHS) program in the United States has been a key federal-funded program in providing comprehensive early childhood services to support low-income families to improve a range of early childhood outcomes that include school readiness, cognitive, language, physical health, and socio-emotional development among young children from infants to age five (Kisker et al., 2002; U.S. Department of Health and Human Services, 2010). From the beginning of the program, engaging fathers and increasing fathers' involvement in HS/EHS has been an important emphasis particularly with increasing recognition of the positive contributions of fathers to parenting and child outcomes (Vogel et al., 2003). Specifically, an integral part of the HS/EHS program has been the promotion of positive parenting practices, the development of healthy parent-child relationships, and supporting fathers[1] involvement and engagement with their children.

However, given the high prevalence of depression among HS/EHS fathers (EHR Research and Evaluation Project, 2006), an area of concern for HS/EHS service providers and practitioners is how fathers' depressive symptoms may adversely affect the quality of their relationships with their children and the possible interventions and engagement

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[1] We use the term "fathers" throughout, but this term encompasses both resident biological fathers as well as a range of male caregivers that serve a father role but may not be a resident biological father.

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strategies that can help to support fathers' depression and promote positive parenting outcomes (National Center on Parent, Family, and Community Engagement, 2013). Recent studies have found that parental self-efficacy can be a protective factor in directly affecting fathers' parenting as well as moderating the effects of depression (Trahan & Shafer, 2019). Using cross-sectional data with a sample of fathers receiving HS/EHS services, this exploratory study aims to understand how fathers' depressive symptoms and PSE are associated with relationship quality with their children and if PSE can have a protective influence for fathers with depression. Though the study is exploratory, it can provide initial knowledge for future studies that can help in the development of HS/EHS services to effectively support fathers' mental health and to create a father-friendly family service approach (Administration for Children and Families, 2018).

Depression is a complex, multiply-determined disorder with genetic and environmental determinants that interact across the life course (Dunn et al., 2015; Repetti et al., 2002). Depression is also one of the common mental health problems in the United States and has been identified as a significant public health issue (McLaughlin, 2011). Further, among adults, depression disproportionately affects parents compared with nonparents with 7.5 million parents experiencing depression in the USA each year (National Research Council and Institute of Medicine, 2009) and about 10% of mothers experiencing depression in the past year as found by a national representative survey of the USA population in 2001–2002 (Ertel et al., 2011). Existing studies indicate that depression among parents may compromise their parenting and adversely affect the development of relationships with their children (e.g., see review by Dix & Meunier, 2009). As parent-child relationships are the cornerstone of child development, understanding the effects of depression on parents' relationships with their children is critical. This is particularly of concern among HS/EHS parents due to the high prevalence of depression identified among these parents. For instance, one multi-site study found nearly 48% of 1275 mothers in EHS were at risk of depression (Administration for Children and Families, 2002). However, much of our current understanding is limited to mothers and maternal depression (Cabrera et al., 2018; Wilson & Durbin, 2010). Less is known about depression among fathers of children in HS/EHS and the impact depressive symptoms have on father-child relationships.

1.1. Depression among fathers of young children

Depressive symptoms may emerge among fathers of young children due to stressors and emotional demands related to nurturing, disciplining, and guiding a child at this young developmental stage (Evenson & Simon, 2010). Several nationally representative studies in the United States indicate that the prevalence of depression is high among fathers of young children, particularly among low-income fathers. Findings from the Early Childhood Longitudinal Study, using the Center for Epidemiological Studies Scale for Depression (CES-D), indicate that approximately 26% of resident biological fathers with young children were mildly or moderately depressed and about 4% were severely depressed (Paulson et al., 2009). In the Fragile Families and Child Wellbeing Study, the prevalence of major depressive episode in the past year among low-income African American fathers was 12% for those with children at both 1 and 3 years of age, and 9% for those with children 5 years of age (Sinkewicz & Lee, 2011).

Several studies have also found a significant prevalence of depression among fathers in HS/EHS programs. One study found that 40% of low-income fathers with infants mostly enrolled in EHS programs were at risk for depression and 23% of fathers had moderate to severe depression when their child was either at 1 month or 14 months (Vogel et al., 2003). Another study using a sample of 80 low-income EHS fathers with 2-year-old children found that the average reported CES-D scores for the fathers were above the clinical cut-off for mild depression (Malin et al., 2012). These studies strongly suggest that depression is experienced by a significant number of low-income fathers with young children in EHS/HS programs.

1.2. Increased risk for depression among low-income fathers of young children

For low-income fathers in EHS/HS, financial demands may create economic pressures that lead to higher emotional distress and increased risk for depression. A study on a community sample of parents with young children in Chicago found that financial strain experienced by low-income parents was associated with higher depressive symptoms (Kingston, 2013). Another study found that high employment needs among low-income fathers from multiple "Responsible Fathering" programs were associated with more depressive symptoms (Fitzgerald et al., 2012).

With the high risk of depression among low-income fathers with young children, one concern is how fathers' depression may affect their parenting functions and relationships with their children since parents' psychological functioning has been found to be a key determinant of parenting outcomes (Belsky, 1984; Finzi-Dottan et al., 2016; Taraban & Shaw, 2018). Existing research, mostly with samples of mothers, strongly shows that mothers' depression is linked with deficits in parenting skills and parent-child relationships (Badovinac et al., 2018; Beck, 1999; Dix & Meunier, 2009). There is a need to understand how depression can affect fathers and their parenting.

1.3. Effects of depression on fathering

Depression impacts parenting and parent-child relationships through disruptions in parents' cognitions, behaviors, and affect (Goodman & Gotlib, 1999; Psychogiou & Parry, 2014). A systematic review of 152 studies found that depression can undermine parent functioning by reducing parents' attention to child input, accentuating negative appraisals of parenting competence, and triggering emotions that are high-negative or low-positive (Dix & Meunier, 2009). Another study found that compared to nondepressed mothers, depressed mothers displayed more negative and less positive interactions with their children (Hops, 1995).

Recent empirical studies and meta-analyses indicate that depression in fathers has comparable effects on fathers' parenting and relationships with their children (Finzi-Dottan et al., 2016; Kane & Garber, 2004; Kopala-Sibley et al., 2017; Lee et al., 2012; Shafer et al., 2019; Sweeney & MacBeth, 2016; Wilson & Durbin, 2010). Wilson and Durbin (2010) concluded from a meta-analysis of 28 studies that there is a significant but small mean effect size for the relationship between paternal depression and decreased positive parenting behaviors (e.g., affectionate & sensitive parenting) and increased negative parenting behaviors (e.g., hostile & coercive parenting). A systematic review of 21 studies found that paternal depression was linked to increased risk of externalizing and internalizing behaviors in children from the age of 2 months to 21 years (Sweeney & MacBeth, 2016). Notably, these associations were stronger among younger children than older children indicating that the developmental period during early childhood is particularly vulnerable to the effects of depression among fathers.

Studies with low-income fathers found that paternal depression can lead to child neglect (Lee et al., 2012), higher incidents of spanking (Davis et al., 2011), and more parent-child conflicts (Finzi-Dottan et al., 2016; Kane & Garber, 2004). Few studies, however, have specifically sought to understand the effects of depression on the parental functioning of fathers in HS/EHS. A study by Vogel et al. (2003) found that depressed EHS fathers of infants spent less time with their children, were less engaged in play and caregiving activities, and showed poorer quality of engagement (e.g., less affect & flexible). This study will fill in the gap in our understanding of HS/EHS fathers by looking at the associations between fathers' depression and relationship quality with their children.

1.4. Parenting self-efficacy

In fatherhood research, the risk and protective framework has been used to understand the determinants of fathers' involvement with their children (Fagan & Lee, 2012). The framework considers the balance of risk and protective (internal and external resources for the protection against risk) factors that interact to determine an individual's ability to function adaptively despite stressful life events (Jenson & Fraser, 2011). Risk factors are conditions that may hinder fathers from having effective engagement and involvement with their children (Fagan & Palkovitz, 2007). Identifying the protective factors that may protect these fathers from the effects of risks is therefore important in developing effective fathering interventions. Parental self-efficacy (PSE) is a parenting-related cognition (Taraban & Shaw, 2018) and has been defined as the "degree to which the parent feels competent and confident in handling child problems" (Fagan et al., 2016, p. 666). We focused on PSE as a protective factor for several reasons. First, there is consistent evidence to support the effects of PSE on parenting outcomes including parenting behaviors, parental stress, and parent-child relationships (Albanese et al., 2019; Jones & Prinz, 2005). Second, PSE is understudied among low-income fathers in the HS/EHS though existing studies with general populations support its role in influencing fathers' involvement (Finzi-Dottan et al., 2016; Trahan, 2018), fathers' warmth (Trahan & Shafer, 2019), and father-child relationship (Dyer et al., 2017). Third, there is evidence that PSE can mitigate the effects of risks on parenting outcomes (Jones & Prinz, 2005; Trahan & Shafer, 2019).

Based on the risk and protective perspective, high levels of PSE are hypothesized to buffer fathers from the negative effects of risks and also increase their engagement and promote positive development of parent-child relationships (Fagan & Lee, 2012; Grote et al., 2007). One study with a general population of fathers found that PSE moderated the relationship between depression and paternal warmth, indicating PSE as a potential protective factor for fathers who are depressed (Trahan & Shafer, 2019). However, the extant literature lacks studies that seek to understand whether PSE can function as a protective factor for low-income fathers in HS/EHS programs.

1.5. Aims of the current study

The current study aims to understand the relationship between fathers' depressive symptoms and the father-child relationship in a sample of low-income fathers from families receiving HS/EHS services. Scholars in fathering research have argued for the shift away from examining the *quantity* of father-child involvement to the *quality* of father-child relationship in understanding the impact of fathers on children's development (Dyer et al., 2017). In this study, we focused on two dimensions of the parent-child relationship quality, father-*child closeness* and *father-child conflict*. Based on Driscoll & Pianta (2011)'s framework for parent-child relationship quality, the father-child relationship can be described by relationship closeness and conflict. Closeness is defined as warmth, affection, and open communication while conflict is defined as behavioral opposition or overt disagreement (Dyer et al., 2017). There is consistent evidence to support the effects of father-child relationship on child well-being (see review by Adamsons & Johnson, 2013).

This study also focused on fathers' PSE as a protective factor. Specifically, we examined if PSE of fathers in HS/EHS can moderate the negative associations between depression and father-child relationship. Fagan and Lee (2012) suggested that the moderating (interactive) model can be used to determine if protective factors can reduce the negative effect of risk factors on positive outcomes. This can be empirically tested by adding interaction terms in the model (Fagan & Lee, 2012). Thus, in this study, we hypothesize that:

Hypothesis 1. Higher depression symptoms will be associated with less father-child closeness and more father-child conflicts.

Hypothesis 2. Higher PSE will be associated with more father-child closeness and less father-child conflicts.

Hypothesis 3. Higher PSE would reduce (i.e., moderate) the negative associations between depression and father-child closeness and conflicts.

Our study aims are novel in that we focused on the quality of father-child relationships in a sample of low-income fathers which is a group that is at risk for depression and lower engagement with their children (especially nonresident fathers; Dyer et al., 2017). We also examined two key predictors of parenting outcomes. Depression is an important determinant of parental functioning (Wilson & Durbin, 2010). PSE, at the same time, is increasingly a key area of focus in parenting interventions (Jones & Prinz, 2005). Therefore, improving our understanding of how PSE can reduce the effects of depression can aid in the development of parenting interventions to support low-income fathers experiencing depression. Examining these issues in the context of HS/EHS programs is relevant because of increasing calls for HS/EHS and other early childhood services to use a father-friendly family service approach (Administration for Children and Families, 2018). Studies showed that higher engagement of fathers in HS/EHS programs are associated with more positive program outcomes for the children (Fagan & Iglesias, 1999). This study can help HS/EHS services to become more aware of fathers in their programs who are experiencing depression that may affect their engagement in programs and to know strategies that can reduce the effects of depression to achieve positive outcomes for children and fathers.

2. Methods

2.1. Data and sample

We analyzed cross-sectional data from a sample of 102 fathers who participated in a larger study examining the effectiveness of a parenting support program. The program was offered by a HS program where the children of the participants attended either HS or EHS preschool. The service boundary of the program covers a county-wide area in North Carolina, USA with a 2013 Rural-Urban Continuum Code 3 designation (i.e., counties in metro areas of fewer than 250,000 population; U.S. Department of Agriculture, 2013). The fathers were randomly assigned to either a treatment group where they were offered a weekly peer support group or a control group where they received usual social services offered by the HS program.

To be eligible for the program, participants had to: (a) be a biological father or a male with a father-figure role, (b) be 18 years or older, (c) speak English, and (d) have at least one child receiving HS or EHS services at the center. Various strategies were used to recruit fathers for the study from April to November 2015. First, printed fliers with study information were displayed in public areas of the HS center and distributed to fathers through the center staff. Second, three fathers whose children were attending HS at the center were recruited to be outreach coordinators to engage fathers during program drop-off and pick-up times for study recruitment. The study team provided detailed explanations of the study and obtained participant consent from recruited fathers. Further information on the parenting support program has been described in Lanier et al. (2019). For this study, we analyzed pre-randomization data collected from all 102 fathers of the study. To recognize the time involved with study participation, fathers received $20 gift cards for the baseline assessments. This study has been approved by the Institutional Review Board at the University of North Carolina at Chapel Hill.

Table 1 shows the sociodemographic characteristics of the study sample. Fathers were on average age 32.42 years (SD = 10.95). The majority of fathers were African American (83%), single in marital status (65%), and had completed high school (46%). Most fathers (57%) earned less than $15,000 in the past year and lived with their child (74%). The focal children were mostly female (60%) and ranged in age

Table 1
Demographic characteristics of the sample (n = 102).

Variable	*M (SD) or n (%)*	
Father age (years)	32.42	10.95
Race		
African American	85	83%
Whites/Others	17	17%
Marital status		
Single	66	65%
Married to child's mother	21	21%
Married other	9	9%
Education level		
No high school	16	16%
High school	47	46%
Some college	39	38%
Income (past 12 months)		
<$5000	32	31%
$5000–$15,000	27	26%
>$15,000	34	33%
Residency with child		
Lives with child	75	74%
Does not live with child	23	23%
Child gender		
Female	61	60%
Male	41	40%
Child age		
<1 year	21	21%
1–5 years	81	79%

Note: Characteristics of the sample before missing data analysis. Some percentages do not add up to 100% due to nonresponse missingness. *M* = mean. *SD* = standard deviation.

from 1 to 5 years (79%).

2.2. Measures

Data were collected using self-report surveys administered to the fathers before randomization. We collected the surveys using offline survey software delivered through a tablet at the HS center. The surveys asked fathers about their relationships with their children, PSE, and depression. Sociodemographic information of the father and child was also collected through the surveys.

Child-parent relationship scale short form (CPRS-SF). The CPRS-SF assesses fathers' self-reports of relationship closeness and conflict with their child (Dyer et al., 2017). Closeness is defined as warmth, affection, and open communication while conflict is defined as behavioral opposition or overt disagreement (Dyer et al., 2017). There are 15 items with seven items related to closeness and eight items related to conflict. The measure was originally designed for parents with children age 12 years and below and was adapted to a sample of low-income nonresident fathers with young children recruited from "Responsible Fatherhood" programs across the United States. Items on the CPRS-SF were scored on a 5-point Likert scale (1 = *definitely does not apply* to 5 = *definitely applies*). Examples of items include "I share an affectionate, warm relationship with my child" (closeness) and "Dealing with my child drains my energy" (conflict). A composite score for each subscale was generated by taking the mean of items. The mean score for closeness was 4.65 ($SD = 0.48$) and for conflict was 1.78 ($SD = 0.69$). A higher score indicates fathers' perceptions of having a closer or more conflictual relationship with their child. Cronbach's alpha (α) for closeness and conflict subscales in this study was 0.78 and 0.82 respectively.

Parenting self-efficacy scale. Fathers' perceptions of their competencies in a role of a parent were measured using a scale developed by the Fatherhood Research and Practice Network (FRPN; Fagan et al., 2016) for children from infancy to age 18. There are seven items where fathers report their level of agreement on a 5-point Likert scale (1 = *strongly disagree* to 5 = *strongly agree*). Examples of items include "I am good at helping my child when he/she is upset or distressed" and "I am good at understanding what my child wants or needs." A composite score for PSE was generated by taking the mean of items ($M = 4.31$, $SD = 1.12$). A higher score indicates higher PSE ($\alpha = 0.98$ in this study).

Depression (CESD-R). The revised version of the Center for Epidemiologic Studies Depression scale was used to measure depression (CESD-R; Eaton et al., 2004). The CESD-R assesses depressive symptoms across 9 groups as described by the American Psychiatric Association Diagnostic and Statistical Manual, fifth edition. These include sadness, loss of interest, appetite, sleep, thinking, tiredness, guilt, movement, and suicidal intention. The CESD-R has 20 items where respondents indicate how often they felt certain ways in the past week based on a 5-point Likert scale (0 = "*not at all or less than 1 day*" to 4 = "*nearly every day for two weeks*"). However, to make the score range of CESD-R equivalent to the original version (CESD; 0–60), the values for the top two responses (i.e., 3 = "*5–7 days*" & 4 = "*Nearly every day for two weeks*") are to be given the same value of three (see Eaton et al., 2004). Item examples include "I lost interest in my usual activities" and "My appetite was poor". All items were summed to obtain a total score for depression ($M = 3.34$, $SD = 5.55$). Higher scores indicating higher depressive symptomatology ($\alpha = 0.70$ in this study).

2.3. Data analysis

We used ordinary least squares linear regression to test the main associations of depression and PSE as well as the moderating effect of PSE on the relationship between depression and father-child relationship outcomes. To test our first and second hypotheses, we ran two main effects models testing the associations of depression and PSE with each of the two father-child relationship outcomes (i.e., closeness and conflict). Following previous studies on fathers' self-efficacy (e.g., Trahan & Shafer, 2019; Lanier et al., 2019), we controlled for sociodemographic variables that are associated with fathering and child outcomes (e.g., father's age, race, & educational level; child's age & child sex).

To test our third hypothesis, we ran an interactional model by adding an interactional term between depression and PSE. We ran the interactional model for each of the two outcomes and subsequent post-hoc tests with Bonferroni corrections made. In all the models conducted, variance inflation factor showed no evidence of multicollinearity. Tests for normality of residuals and test for heteroskedasticity in the variance of the residuals using Breusch-Pagan test and White's test indicated that there was no evidence of skewness and heteroskedasticity in the residuals. We further examined the data for outliers and data points that had high leverage influence on the parameters using studentized residuals, Cook's D, and leverage scores. Only one case was identified to be an outlier and had high leverage; this case was excluded from the regression analysis, thus reducing the sample size to 101. Data missingness ranged from 3% to 7% across the variables. Little's test indicated that missing data were missing completely at random ($p = .23$). Therefore, to retain the sample size, we used multivariate imputation by chained equations (Graham, 2008) to impute possible values for missing data in 30 new datasets. 48 Analyses were subsequently conducted on each imputed data set and parameters were appropriately aggregated. Post-hoc analysis indicated that the current study was able to detect medium effect sizes ($d = 0.56$) given a power at the 0.80 level and alpha at the 0.05 level. All analyses in this study were completed using Stata version 14.2.

3. Results

To test research hypotheses 1 and 2, we estimated two main effect models to examine the associations of depression and PSE with two relationship outcomes (i.e., father-child closeness and conflict). The results for the two models (see Table 2) show that paternal depression was significantly associated with relationship closeness ($B = -0.37$, $p <$

Table 2
Main effects models (n = 102).

	Closeness			Conflict		
	B		SE	*B*		SE
Depression	−0.37	***	0.01	0.28	*	0.02
Parental Self-efficacy	0.02		0.03	−0.07		0.05
Father age	−0.20	*	0.003	0.24		0.01
Father race (African Americans)						
Whites/Others	−0.27	**	0.11	0.09		0.21
Residency with chid						
No, does not stay with child	−0.17		0.13	0.24	*	0.20
Educational level						
High school	−0.39	***	0.10	0.01		0.26
Some college	−0.32	*	0.12	−0.04		0.27
Father's income						
$5000–$15,000	0.08		0.12	0.00		0.17
>$15,000	0.15		0.10	−0.14		0.16
Father Marital Status						
Married to child's mother	−0.16		0.11	0.06		0.16
Married other	0.18		0.15	−0.18		0.26
Child age						
1–5 years	0.24	*	0.12	0.09		0.13
Child gender						
Male	0.10		0.08	−0.03		0.15
Adjusted R^2	0.32			0.20		

Note: * $p < .05$, ** $p < .01$, *** $p < .001$.
B = betas or standardized regression coefficients.

Table 3
Interactional models (n = 102).

	Closeness			Conflict		
	B		SE	*B*		SE
Depression	−1.23	**	0.03	1.30	***	0.05
Parental Self-efficacy	−0.08		0.03	0.05		0.06
Depression × Self-efficacy	0.88	*	0.01	−1.05	*	0.01
Father age	−0.19	*	0.003	0.23		0.01
Father race						
Whites/Others	−0.29	***	0.10	0.11		0.20
Residency with child						
No, does not stay with child	−0.17		0.13	0.24	*	0.19
Educational level						
High school	−0.38	***	0.10	0.01		0.25
Some college	−0.34	**	0.12	−0.01		0.27
Father's income						
$5000–$15,000	0.08		0.12	−0.0002		0.17
>$15,000	0.14		0.10	−0.14		0.16
Father Marital Status						
Married to child's mother	−0.17		0.10	0.07		0.16
Married other	0.14		0.15	−0.14		0.27
Child age						
1–5 years	0.27	*	0.12	0.06		0.14
Child gender						
Male	0.07		0.07	0.005		0.15
Adjusted R^2	0.35			0.24		

Note: * $p < .05$, ** $p < .01$, *** $p < .001$.
B = Betas or standardized regression coefficients.

.001) and conflict ($B = 0.28$, $p < .05$) above and beyond the associations of PSE and sociodemographic covariates. Specifically, one standard deviation increase in depressive symptoms was associated with a 0.37 standard deviation decrease in relationship closeness and 0.28 standard deviation increase in conflicts. PSE was not statistically associated with any of the relationship outcomes after controlling for the associations of depression and other covariates at their mean scores. Sociodemographic covariates were associated with different outcomes. For relationship closeness, older fathers, fathers of White/Other races, fathers with an educational level at high school or higher, and fathers with young children reported significantly lesser closeness in relationships with their children. For relationship conflict, nonresident fathers reported significantly more conflicts with their children. The main effect model for relationship closeness explained 33% of the variance in the outcome ($F(13,84.9) = 3.36$, $p < .001$); the model for relationship conflict explained 20% of the variance ($F(13,84.9) = 1.98$, $p < .05$).

To test hypothesis 3, we estimated an interaction model for each outcome by including an interactional term between depression and PSE while controlling for similar sociodemographic covariates used in the main effect models. Results in Table 3 indicate that there is a significant interactional influence for PSE on the associations of depression with both relationship outcomes. Specifically, PSE significantly moderated the relationship between depression and (a) relationship closeness ($B = 0.88$, $p < .05$) and (b) conflict ($B = -1.05$, $p < .05$). In addition, the standardized regression coefficients for the moderating term are greater in magnitude for conflict than for closeness indicating that PSE has a stronger moderating influence for conflict outcome than for closeness.

The moderation influences are visually illustrated in Fig. 1 and Fig. 2 for relationship closeness and conflict respectively. In both plots, we standardized PSE and both relationships outcomes to a mean of zero and standard deviation of one. For depression, we standardized it to a mean of 16 (the cut-off for clinical depression; see Eaton et al., 2004). Following Trahan & Shafer (2019), we then plotted the relationships between standardized depression and standardized relationship outcome when (a) PSE is low (1 standard deviation below mean), (b) PSE is at average (at mean), and (c) PSE is high (1 standard deviation above mean).

In Fig. 1, when depression was low (i.e., one standard deviation below the cut-off for clinical depression), fathers with low PSE reported relationships that were about 0.4 standard deviations less close than fathers with average PSE. Fathers with high PSE, on the other hand, had relationships that were 0.4 standard deviations closer than fathers with average PSE. As shown in Fig. 1, these differences with average PSE widens when depression scores increased. When depression was high (1 standard deviation above the cut-off), fathers with low PSE had relationships that were about 0.8 standard deviations less close than fathers with average PSE. Fathers with high PSE, on the other hand, had relationships that were 0.8 standard deviations closer than fathers with average PSE. Subsequent tests showed that low PSE fathers with high depression had significantly less close relationships with their children than low PSE fathers with low depression ($B = -0.55$, $p < .001$). Results were similar for average PSE fathers ($B = -0.31$, $p < .01$). However, for fathers with high PSE, there were no significant differences in closeness scores between fathers with high depression and low depression ($B = -0.08$, $p = .70$).

The moderating influence of PSE on the relationship between depression and relationship conflict is similarly illustrated in Fig. 2. When depression was low, fathers with low PSE reported more conflictual relationships (0.5 standard deviations more) than fathers with average PSE. Fathers with high PSE, on the other hand, had less conflictual relationships (0.5 standard deviations less) than fathers with average PSE. As shown in Fig. 2, these differences with average PSE widens when depression scores increased. When depression was high, fathers with low PSE reported more relationship conflicts (1 standard deviation more) than fathers with average PSE. Fathers with high PSE, on the other hand, had less conflicts (1 standard deviation lower) than fathers with average PSE. Subsequent tests showed that low PSE fathers with high depression had significantly higher conflict scores than low PSE fathers with low depression ($B = 0.46$, $p < .001$). However, there were no significant differences in conflict scores among fathers with high depression and low depression among those with average PSE ($B = 0.18$, $p = .12$) as well as those with high PSE ($B = -0.10$, $p = .61$).

The interactional model for closeness explained 35% of the variance

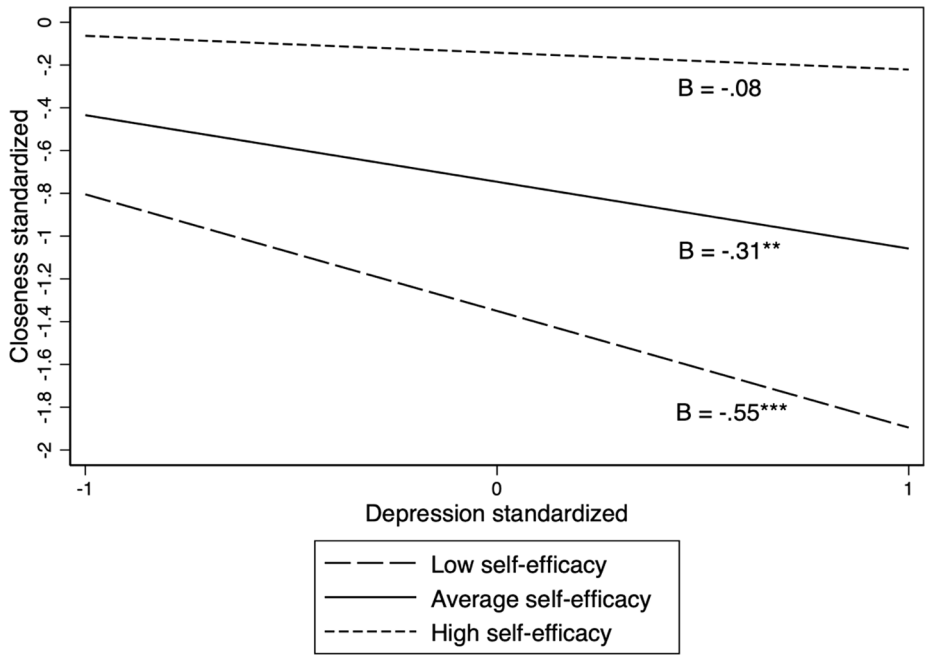

Fig. 1. Effects of parental self-efficacy on the relationship between depression and closeness.

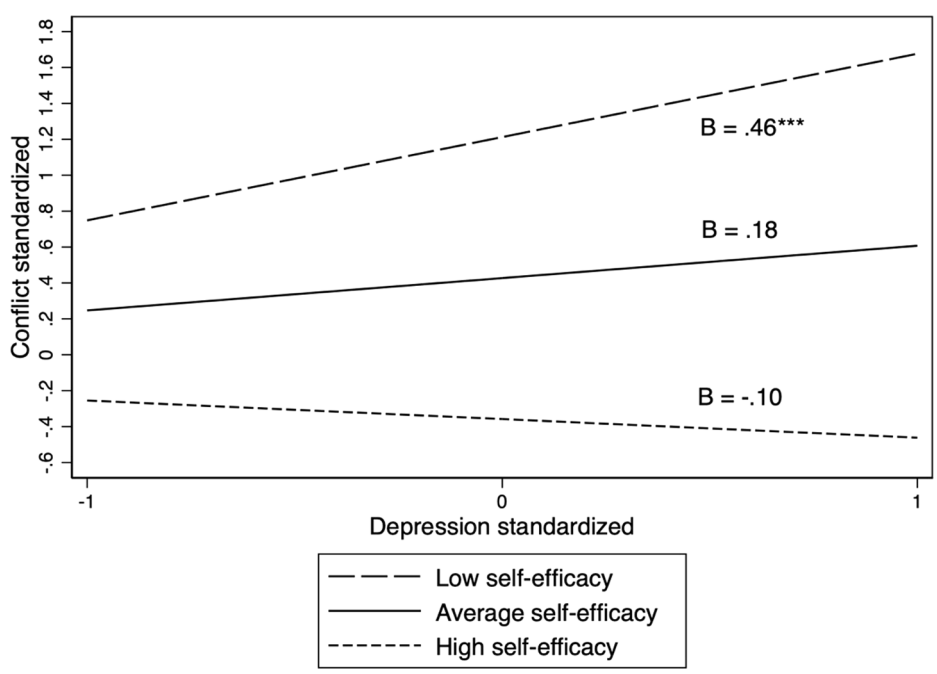

Fig. 2. Effects of parental self-efficacy on the relationship between depression and conflict.

in the outcome ($F(14,83.8) = 4.55$, $p < .001$) while the interactional model for conflict explained 24% of the variance in the outcome ($F(14,83.7) = 7.36$, $p < .001$). Relationships between sociodemographic covariates and the outcomes were similar to those found in the main effects models—fathers who were older, of White/Other races, had educational levels at high school or higher, or had younger children were more likely to report lower closeness in their relationships with their children.

4. Discussion

The goals of this study were (a) to understand how depression and PSE are associated with relationship closeness and conflict among a sample of low-income HS/EHS fathers with young children, and (b) to examine if PSE would moderate the associations between depression and father-child relationship. Improving our knowledge in these areas can help improve strategies and interventions to support fathers in HS/EHS program.

The results showed that paternal depression among low-income HS/EHS fathers with young children was associated with reduced father-child closeness and increased father-child conflict. These findings are in line with existing studies with broader populations of fathers (Shafer et al., 2019; Trahan & Shafer, 2019). Since existing studies had mostly focused on mothers, or sampled fathers broadly, our study adds to this

body of knowledge by examining depression among a more homogenous sample of low-income fathers with young children in HS/EHS. The associations between depressive symptoms and poorer relationship outcomes found in this study were statistically significant even after controlling for variations in PSE and sociodemographic characteristics such as fathers' age, child gender, residency status, and family income. This result indicates the importance of assessing depression among HS/EHS fathers with young children where the risk of depression could be higher as a result of challenging economic pressures and emotional demands associated with parenting children at this developmental stage. Depression can affect fathers' parental functioning through ways such as reduced empathic understanding towards their child and increased negative displays of affect which in turn compromises the quality of father-child relationships (Psychogiou & Parry, 2014).

The second key finding of this study is that fathers' PSE moderated the associations between depression and their relationships with their children. Specifically, we found that for fathers with a higher sense of efficacy in their parenting role, the negative associations of depression with relationship closeness and conflicts are smaller or negligible compared to fathers who do not feel competent about parenting. This result lends empirical support to our conceptualization of PSE as a protective factor that moderates (buffers) the adverse effects of depression on the father-child relationship.

Depression could undermine fathers' attention to their child's goals and needs, increase self-ruminations, initiate negative affect, and decrease their satisfaction in being a father (Psychogiou & Parry, 2014). Thus, for depressed fathers who can maintain a sense of confidence in their parenting abilities, PSE helps them to sustain a level of parental functioning that is still oriented and adaptive to the goals and developmental needs of the child (Jones & Prinz, 2005). A higher sense of PSE can help depressed fathers to remain committed to their identity role as a father or maintain a high degree of role satisfaction (Trahan & Cheung, 2012). Furthermore, fathers who are efficacious may be more likely to cope by seeking social support from others to aid them in parenting tasks (Finzi-Dottan et al., 2016).

Contrary to what we have expected, fathers' PSE was not associated with relationship closeness or conflicts with their children after the associations of depression were controlled for. Existing studies mostly conducted with mothers have found direct effects of PSE on parent-child relationships (Jones & Prinz, 2005). Trahan (2018) found that fathers' PSE predicted higher father involvement; however, the study did not look at relationship quality and it used a broad sample of fathers across different socioeconomic status. Though further investigations are warranted, our study suggests that first, paternal depression has a significant impact on low-income HS/EHS fathers' relationships with their children such that after accounting for the effects of depression, PSE could not explain any remaining variations in the outcomes. Second, when PSE and depression among fathers are studied together in an interactional model, the influence of PSE manifest strongly as a protective factor that moderates the effects of risk factors. Thus, this indicates that the risk and protective framework is a useful model for future studies to examine the interactions among hypothesized risk and protective factors that can influence father involvement outcomes (Fagan & Lee, 2012).

The results also indicate that PSE had a stronger moderating influence on the effects of depression on conflict than for closeness. Significant differences in closeness score persist between fathers of high and low depression at both low and average levels of PSE; for conflict, however, these differences only persist for fathers at low PSE. In other words, a higher level of PSE was needed to counteract the associations of depression with relationship closeness than for conflicts. One possible explanation is that fathers in this study may be less receptive to contemporary fatherhood norms associated with warmth and nurturing engagement with their children and a higher level of PSE may be needed to counteract the effects of depression. In a study that used a gendered response framework to understand the effects of paternal depression on fathers' involvement, Shafer et al. (2019) found that depression affected fathers' warmth and harsh parenting depending on their adherence to traditional or non-traditional masculine norms. The former may inhibit fathers' nurturing behaviors and emotional engagement associated with contemporary fatherhood norms but encourage harsh parenting practices. Further studies are certainly needed to investigate these differences in the moderating effects of PSE in relation to relationship outcomes.

Finally, although not a primary focus of the study, relationship closeness and conflict were related to several sociodemographic characteristics. Younger fathers, African American fathers, fathers with older children, and those with an education level lower than high school rated themselves to have higher levels of closeness with their children. However, for relationship conflict, only fathers' residency with child predicted differences. Previous studies indicate that when fathers lose their residency status, there are major qualitative changes in their relationship with their children (Dyer et al., 2017) such as higher father-child conflicts as indicated by this study. Relationship closeness and conflict are constructs that represent unique aspects of father-child relationships (Dyer et al., 2017) and this study shows that they have different associations with sociodemographic factors. However, these sociodemographic factors could also be proxies for other unobserved factors that are determinants of parenting such as culture, parenting norms, social support, and resources.

4.1. Limitations

This study has several limitations that should be considered in the interpretation of the findings. First, we used cross-sectional data that could not help us to determine if PSE and depression had causal effects on the outcomes over time. For instance, fathers who have more conflictual relationships could be subsequently more depressed. Second, we did not include other unobserved factors such as fathers' role commitment and dyadic interactions with mothers (e.g., Chung et al., 2020) that could influence the outcomes. Mothers' parenting and depressive symptoms are also not controlled for because this information was not collected. Accounting simultaneously for the relative influence of mothers' mental health can better evaluate the effects of fathers' PSE and depressive symptoms on their parenting and children outcomes especially for the co-resident fathers (see Meadows et al., 2007 for discussion on dual parents' internalizing psychopathology). Third, we focused only on relationship closeness and conflict and did not include other fathering outcomes. Thus, we are unsure if the relationships found in this study could apply to other forms of father involvement and engagement. Fourth, fathers in the sample are concurrently receiving HS/EHS services which may affect variations in fathers' parenting and outcomes that are different from fathers who are not receiving services. Furthermore, this sample of low-income fathers with young children were sampled from a specific geographic location in the United States. Although this helps us to better study our research questions in a homogenous group, it limits our generalizability to other types of fathers who are not receiving service, from other geographical regions in the USA or other countries, fathers who are not low-income, or fathers in the wider HS/EHS population. Fifth, while we had accounted for contextual factors such as poverty, relationship status with mother, and residency with child, other key contextual factors such as fathers' social networks, social support and work environment that can influence the quality of fathers' parenting are not accounted for in this study (Castillo & Sarver, 2012). Cabrera et al. (2014)'s expanded model of father involvement emphasized contextual factors that influence how fathers parent. Sixth, all the scales were self-reported by fathers and no alternative sources of data were used to verify fathers' report. Lastly, our sample size is small and we were unable to include additional interaction terms among our independent variables due to the lack of power.

Future studies could use longitudinal data, incorporate mothers' and child's reporting of outcomes, and account for the impact of mothers'

co-parenting to better examine the effects of PSE and depression on parenting outcomes. From a systemic perspective, children's behaviors can also have an impact on parental responses (i.e. a child-driven pathway). Future studies could consider the effects of child's behaviors on parenting practices or PSE in the context of fathers' depression (e.g., Glatz et al., 2017). To confirm the results of this exploratory study, future studies should recruit fathers from large, representative samples of parents in the HS/EHS population.

4.2. Implications

Nonetheless, this study suggests the potential risk of depression on fathers' relationships with their children and has several implications for HE/EHS programs. While there have been calls for HS/EHS to increase the involvement of fathers with their children and improve father-child relationships, efforts to do it should not overlook the effects of fathers' depression. HS/EHS services should give attention to providing mental health assessment and treatment to help fathers cope with their depressive symptoms. Given HS/EHS's emphasis on creating a father-friendly family service approach (Administration for Children and Families, 2018), a key consideration is how workers engage fathers experiencing depression in their day-to-day practice. For instance, HS/EHS workers could consider increasing their affirmations of fathers' efforts in parenting and showing sensitivity to fathers' limitations and circumstances. This may include putting into place protocols for workers to make external referrals for fathers to get services and help for their problems. This also means that HS/EHS programs need to work with other services to build a system that encompasses a wide range of programs and services that serves both the child and the adult (Azzi-Lessing, 2011). While these programs may include parenting and mental health services, it may also include substance abuse, family self-efficiency programs, and many others that vulnerable families such as those in HS/EHS may need.

This study also highlights the potential role of PSE as a protective factor in counteracting the effects of depression. Existing reviews of PSE interventions found that group-based interventions for parents with children of age 6 years and below are efficient, cost-effective, and effective in influencing parenting behavior (Wittkowski et al., 2016). There are some evidence that single-session interventions focusing on specific parenting situations (e.g. mealtimes or shopping) can be just as effective as multiple sessions in increasing PSE of parents (see Wittkowski et al., 2016). Effective PSE interventions were, however, found to be developed based on the principles of Bandura's theory of self-efficacy. Given its time and cost-effectiveness, incorporating group-based PSE interventions to improve fathers' PSE can be beneficial for HS/EHS programs that aim to promote positive parenting practices and develop healthy parent-child relationships for fathers with depression. Moreover, existing studies with HS/EHS fathers found that service approaches that merely emphasize father involvement are not as effective at improving parenting and child outcomes as those approaches that focus on improving fathers' parenting competencies such as PSE (Fagan & Iglesias, 1999).

CRediT authorship contribution statement

Gerard Chung: Conceptualization, Writing - original draft, Writing - review & editing, Formal analysis. **Quinton Smith:** Investigation, Writing - review & editing, Project administration. **Joseph Frey:** Investigation, Writing - review & editing, Project administration. **Paul Lanier:** Writing - original draft, Investigation, Writing - review & editing, Project administration, Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.childyouth.2020.105758.

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